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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: RE26-SE-025

1. Topic (12 words max)

AI-Integrated Multi-Modal Academic Evaluation, Performance Intelligence, and Automated Viva Assessment System

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	RE26-SE-025
Year	2026

5. Brief description of the research problem

The increasing scale of higher education, continuous assessment practices, and online learning environments has exposed significant limitations in traditional academic evaluation methods. Manual assessment of written exams, assignments, diagrams, and viva examinations is highly time-consuming, prone to inconsistency, and difficult to scale. This problem primarily affects educational institutions, lecturers, examiners, and students, where evaluators face excessive workloads and students experience delayed feedback and potential subjectivity in grading. The future users of the proposed solution include universities, examination boards, and online education platforms seeking reliable and efficient evaluation systems. [1]

In addition, human-centered factors such as examiner fatigue, cognitive load, and emotional state significantly influence the reliability of manual grading processes. As evaluators assess large volumes of scripts over extended periods, their attention, judgment consistency, and tolerance levels may fluctuate, leading to unintentional variability in scoring quality across early and later assessments. This temporal inconsistency introduces grading bias that is unrelated to student performance but rather examining exhaustion and mood drift, thereby undermining the objectivity and fairness of academic evaluation. [2]

The impact of the problem spans multiple dimensions. Socially, inconsistent grading and delayed result processing reduce student trust in assessment fairness and hinder learning improvement. Technically, existing automated assessment tools are limited in scope, often relying on keyword-based evaluation and lacking the ability to interpret handwritten content, diagrams, formulas, and oral responses. Economically, institutions incur high operational costs due to extensive human involvement in marking large volumes of academic work. From an ethical and legal

perspective, subjective evaluation and unconscious examiner bias raise concerns regarding fairness, transparency, and accountability in high-stakes academic assessments. [3]

This research is positioned within the Artificial Intelligence and Data Science research cluster, with strong relevance to the Education Technology (EdTech) and Intelligent Decision Support Systems industry verticals. The project aims to address the identified problem by designing an AI-integrated, multi-modal academic evaluation framework capable of analyzing textual, visual, and verbal student responses using a unified evaluation engine.

By introducing an explainable, assistive AI-based assessment approach, the proposed research seeks to improve evaluation consistency, reduce examiner workload, and enhance the overall reliability of academic assessment processes while maintaining human oversight and institutional control.

6. Brief description of Existing Research and Systems

Several academic assessment and examination support systems are currently in use across educational institutions and online learning platforms. Learning Management Systems (LMS) such as Moodle, Blackboard, and Canvas provide basic automated assessment features, primarily supporting multiple-choice questions, simple quizzes, and rule-based grading. These systems offer limited automation for subjective assessments such as essays, handwritten examinations, diagrams, and viva evaluations, often requiring extensive manual intervention.

Automated essay scoring systems, including tools based on traditional Natural Language Processing techniques, focus largely on keyword matching, surface-level semantic similarity, and statistical language features. While these systems improve grading speed for large-scale standardized tests, they lack the ability to evaluate reasoning quality, conceptual understanding, and logical coherence. Diagram and formula evaluation tools exist mainly within specialized domains such as mathematics or engineering education, but they operate in isolation and are not integrated into a unified academic evaluation framework.

Online proctoring and viva assessment platforms primarily focus on identity verification, cheating detection, and session recording. These systems provide minimal support for structured evaluation of oral responses, confidence levels, or conceptual clarity, leaving final judgment entirely to human examiners. Overall, existing products function as fragmented solutions, addressing isolated aspects of academic assessment rather than providing a comprehensive, scalable evaluation system.

Academic research in automated assessment has progressed significantly with the advancement of Artificial Intelligence and Data Science. Natural Language Processing techniques have been widely explored for automated essay grading, including semantic similarity modeling, topic coverage analysis, and machine learning-based scoring methods. Recent studies have introduced deep learning models such as transformers to improve contextual understanding in textual evaluation.

Based on the review of existing systems and research, several critical gaps have been identified

- Lack of a unified multi-modal evaluation framework capable of assessing text, handwriting, diagrams, formulas, and viva responses within a single system.
- Overreliance on keyword-based or surface-level semantic analysis, resulting in poor reasoning and conceptual understanding assessment.
- Inability to address examiner fatigue, temporal grading inconsistency, and subjective bias inherent in manual evaluation processes.
- Limited explainability of AI-generated scores, reducing transparency and trust in automated assessment.
- Absence of effective human-in-the-loop mechanisms for controlled AI-assisted grading.
- Poor scalability for large-scale academic evaluations involving diverse assessment formats.
- Insufficient use of predictive analytics to identify learning gaps and support academic decision-making.
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7. Brief description of the solution's nature

The proposed solution is an AI-integrated, multi-modal academic evaluation system designed to overcome the limitations of traditional manual assessment and existing automated grading approaches. Conventional evaluation methods are often time-intensive, inconsistent, and constrained to single-mode analysis such as keyword-based text matching. In contrast, the proposed system introduces a unified, context-aware evaluation framework capable of assessing diverse academic response formats within a single intelligent platform.

The system supports the evaluation of typed textual answers, handwritten examination scripts, diagrams, mathematical expressions, and oral responses from online viva examinations. By integrating multiple artificial intelligence techniques, including natural language processing, computer vision, speech analysis, and predictive analytics, the framework ensures scalable, consistent, and transparent academic assessment while maintaining human oversight.

This research directly addresses previously identified challenges such as subjective grading, examiner fatigue, limited automation, lack of reasoning-level understanding,

and scalability constraints. The system evaluates not only answer correctness but also conceptual depth, logical coherence, reasoning quality, and structural accuracy. Additionally, explainable AI mechanisms are incorporated to provide justification for all assessment decisions, ensuring accountability and trustworthiness.

System Perspective

Input

- The system accepts a wide range of academic data sources.
- Typed answers, essays, assignments, and academic reports.
- Scanned handwritten examination scripts and diagrams captured via a mobile application.
- Mathematical formulas, equations, symbols, and diagrammatic representations.
- Recorded online viva examination sessions, including video and audio streams.
- Examiner-defined marking schemes, grading rubrics, and historical academic records

Processing Architecture

Multi-Modal Data Acquisition and Preprocessing

Scanned answer scripts undergo advanced OCR and handwriting recognition, while diagrams are extracted and converted into structured representations. Audio and video data from viva examinations are processed to extract speech transcripts, vocal characteristics, facial expressions, and behavioral cues.

Unified AI-Based Evaluation Engine

Textual responses are evaluated using semantic similarity analysis, knowledge coverage estimation, misconception detection, and reasoning-pattern alignment with marking schemes. Diagrams and mathematical content are assessed using structural, graph-based, and symbolic matching techniques against expected rubric-defined solutions. Viva responses are analyzed through linguistic, vocal, and facial expression features to assess content accuracy, clarity, confidence, and conceptual understanding.

Predictive Analytics and Insight Generation

Aggregated evaluation data is analyzed to identify learning gaps, weak concepts, and question-level performance trends. A predictive analytics module leverages historical assessment data to forecast student performance and generate potential future examination questions.

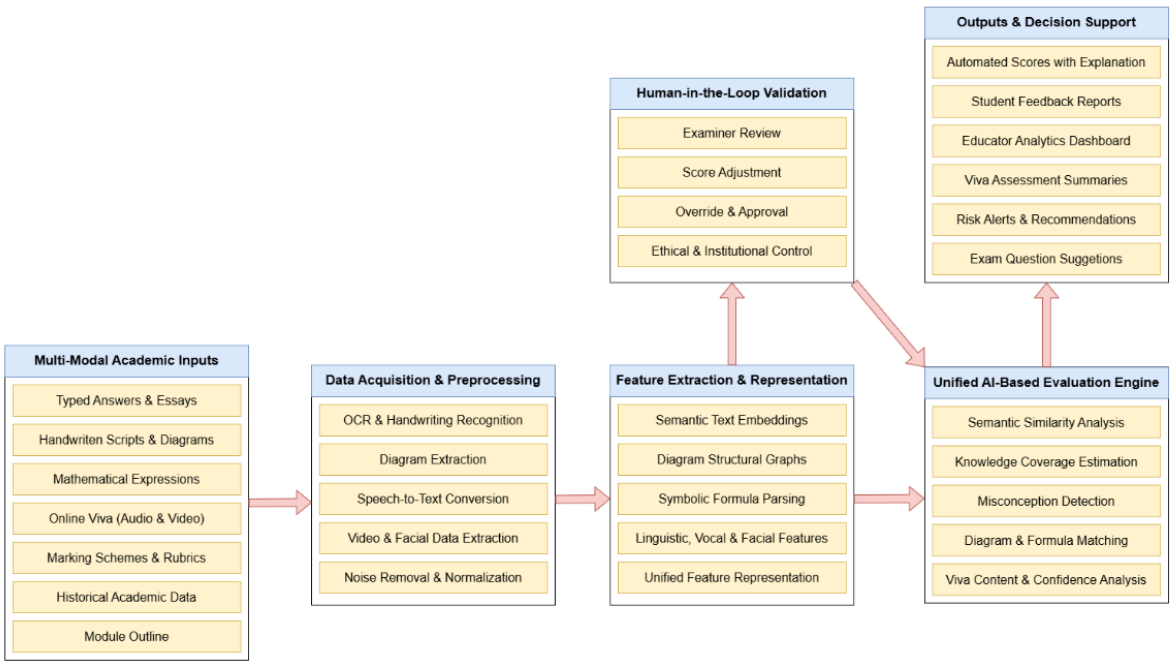
Explainability and Human-in-the-Loop Validation

Each AI-generated evaluation is accompanied by confidence scores, justification metrics, and explanatory feedback. Educators retain full authority to review, modify, or override assessment outcomes, ensuring ethical compliance and institutional control.

Output

- Automated assessment scores with detailed explanations
- Structured, personalized feedback reports for students
- Performance analytics dashboards for lecturers and institutions
- Predicted learning risks and recommended focus areas
- Comprehensive viva examination summaries with confidence indicators

Conceptual Diagram



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data.

The successful development of the proposed AI integrated multi modal academic evaluation system requires expertise spanning artificial intelligence, educational assessment, computer vision, natural language processing, data analytics, and human computer interaction. The project team can acquire the required specialized domain expertise through a combination of structured academic study, guided supervision, and targeted self-learning. Foundational knowledge in educational evaluation principles, marking rubrics, and assessment fairness can be gained through reviewing academic literature, university assessment guidelines, and consultation with lecturers and examiners. Regular feedback sessions with academic supervisors and domain experts will further ensure alignment between AI-driven evaluation outcomes and real-world academic assessment standards.

The technical knowledge and skills required for this project include machine learning, deep learning, multi-modal data processing, and system integration. Team members can develop these skills through online courses, research papers, and hands-on experimentation with frameworks such as TensorFlow, PyTorch, OpenCV, and NLP libraries. Prototyping individual modules such as OCR pipelines, semantic similarity models, and audio video analysis components allows members to progressively build competence while validating design choices. Collaborative development practices, code reviews, and version control further support knowledge sharing within the team.

The data required for the project includes written and handwritten student answers, scanned diagrams, academic rubrics, and recorded viva style oral responses. Data collection can be conducted through multiple approaches. Publicly available educational datasets, synthetic data generation, and anonymized historical academic materials can be used for training and testing. Handwritten samples and diagrammatic answers can be created by volunteers using predefined templates to simulate real exam conditions. For viva analysis, mock interview sessions with

informed participants can be recorded in controlled environments. All collected data will be anonymized, labeled where necessary, and stored securely to prevent unauthorized access.

This project does require ethical clearance, as it involves human-generated data such as written responses, voice recordings, and video recordings. Ethical approval ensures that participant consent, data privacy, confidentiality, and responsible AI usage are properly addressed. Participants will be informed about the purpose of data collection, usage scope, and their right to withdraw. No real academic decisions will be made solely based on the system outputs during the research phase, ensuring that the system remains an assistive tool rather than a decision-making authority.

9. Objectives and Novelty

Main Objective The system is an AI-powered multi-modal academic evaluation platform that auto-grades written and digital answers, scans handwritten papers, analyzes diagrams, predicts student performance, generates future exam questions, and evaluates online viva sessions through video, voice, and expression analysis.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22262554	Predictive Learning Analytics and Intelligent Question Generator	**Deep Question-Level Analytics** - Which questions were most misunderstood, which cognitive skills **AI-Powered Weak-Point Diagnosis Engine** - Weak topics, Weak question types, Conceptual gaps **Future Exam Question Predictor** - Studying past exam patterns, identifying topics repeatedly emphasized by lecturers, noticing concept clusters frequently assessed, Predicting likely question structures	Analyze answers using an intelligent engine, and generate personalized, weak-point–focused questions to directly address identified learning gaps.
IT22134776	Intelligent Mobile Scanning, Diagram Extraction, and Multi-Modal	**Auto-Capture + Page-Turn Detection** - Uses motion sensors + camera frames to detect page flips and auto-capture images.	Rubric-aware diagram extraction, understanding, and automated grading using diagram analysis.

	Answer Analysis Module	**Advanced Multi-Modal OCR** (Text + Handwriting + Diagrams) - Handwriting OCR using Transformer-based handwriting models, convert diagrams into structured knowledge graphs, Compare with expected diagram structure from rubric **Multi-Modal Answer Analysis** - Textual content is semantically matched with expected answers, Diagrams are structurally compared using graph matching algorithms, Formulas, equations, and mathematical symbols are parsed and evaluated	
IT22557056	AI-Enhanced Online Viva Evaluation Module	**Voice Analysis** - Speech clarity, Confidence level, Hesitation patterns, Pitch stability, Truthfulness/Authenticity estimation using micro-pause behavior **Facial Expression Analysis** - Eye contact, Stress indicators, Facial emotion recognition, Confidence & honesty estimation using micro-expressions **Content Understanding** - Accuracy of technical answers, Depth of explanation, Logical structure, Ability to justify decisions	A multi-modal AI framework that evaluates viva performance using speech, facial, and content analysis with explainable feedback.

		Viva Panel Feedback Automation - Summary report for the panel, Score suggestions with justification, Confidence score for each evaluation	
IT22239198	A unified AI engine for fair, human like evaluation of exams, essays, assignments, and reports.	**Interpret grading rubrics** – Check reasoning, follow marking schemes, and detect errors or irrelevance. **NLP evaluation** – Analyze content for coverage, misconceptions, and semantic similarity. **Check structure and originality** – Ensure proper format, concepts, and authenticity. **Unified framework** – Combine all modules for consistent assessment.	**Unified evaluation** – Assess multiple academic formats with reasoning-focused grading. **Context-aware checks** – Apply penalties and detect misconceptions intelligently. **Authenticity verification** – Ensure originality during evaluation.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Sanjeevi	Chandrasiri	Sanjeevi
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				